

- (a) New World monkeys, such as spider monkeys (shown here), squirrel monkeys, and capuchins, have a prehensile tail (one adapted for grasping) and nostrils that open to the sides.



- (b) Old World monkeys lack a prehensile tail, and their nostrils open downward. This group includes macaques (shown here), mandrills, baboons, and rhesus monkeys.

▲ **Figure 34.45** New World monkeys and Old World monkeys.

about 25–30 million years ago. Today, nonhuman apes are found exclusively in tropical regions of the Old World. With the exception of gibbons, living apes are larger than either New or Old World monkeys. All living apes have relatively long arms, short legs, and no tail. Although all nonhuman apes spend time in trees, only gibbons and orangutans are primarily arboreal. Social organization varies among the apes; gorillas and chimpanzees are highly social. Finally, compared to other primates, apes have a larger brain in proportion to their body size, and their behavior is more flexible. These two characteristics are especially prominent in our next group, the hominins.

#### CONCEPT CHECK 34.6

1. Contrast monotremes, marsupials, and eutherians in terms of how they bear young.
2. Identify at least five derived traits of primates.
3. **MAKE CONNECTIONS** Develop a hypothesis to explain why the diversity of mammals increased in the Cenozoic. Your explanation should consider mammalian adaptations as well as factors such as mass extinctions and continental drift (review Concept 25.4).

For suggested answers, see Appendix A.

#### ▼ **Figure 34.46** Nonhuman apes.

- (a) Gibbons (18 species), such as this Muller's gibbon, are found only in south-eastern Asia. Their very long arms and fingers are adaptations for brachiating (swinging by the arms from branch to branch).



- (b) Orangutans (3 species) are shy apes that live in the rain forests of Sumatra and Borneo. Named in 1917, this species (*Pongo tapanuliensis*) is the first great ape discovered in nearly 100 years.



- (c) Gorillas (2 species) are the largest apes; some males are almost 2 m tall and weigh about 200 kg. Found only in Africa, these herbivores usually live in groups of up to about 20 individuals.



- (d) Chimpanzees live in tropical Africa. They feed and sleep in trees but also spend a great deal of time on the ground. Chimpanzees are intelligent, communicative, and social.



- (e) Bonobos are in the same genus (*Pan*) as chimpanzees but are smaller. They survive today only in the African nation of Congo.

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